TDT4259 – Applied Data Science

Lecture 2: Bringing value from datasets

Nisha Dalal
Adj. Associate Professor


nisha.dalal@ntnu.no

Agenda

- **10:15 – 10:20 Introduction**
- **10:20 – 10:40 Grading criteria and suggested template**
- **10:40 –11:00 NTNU**
- **11:00- 11:15 SINTEF Energy**
- **11:15- 11:30 Break**
- **11:30- 11:45 SINTEF Ocean**
- **11:45 – 12:00 ANEO AS**

Reference groups and feedback

We are looking for 5-8 students to comprise the reference group. The purpose of the reference group is to provide constructive feedback about the course through an ongoing open dialogue with other students throughout the semester.
You can read more about task of the reference group in this link.

If you want to sign up to be a member of the reference group, use this link. Also available in Canvas (Modules -> Important information and links)

A survey will be sent out to all to evaluate the course during the last week.

Project (Group)

You will work in groups of 5-6 students.

Fill the form to register as a group. A **separate link for individuals** to sign up for **the group project.** Links also present in Canvas (Modules -> Important information and links)

IMPORTANT! Only one entry is needed per team.

IMPORTANT! Deadline is 16th September!

Don't make groups in Canvas!!

The final deliverable will be a pdf report.

Contributes to **60% of your final grade!**

Project (Group)

TAKE YOUR TIME and form groups by yourself.

If signed up individually, don't worry! Just sign up as a group and we will discard your previous registration.

If you change groups, register again with new details. We will discard the previous registration.

Project (Group): Task Description

How to solve a business problem using data?

How to create value from data?

How to make data-driven decisions?

Grading criteria

1. **Problem definition & business understanding (15%)**
2. **Data understanding & background (15%)**
3. **Methods, analysis & technical evaluation (20%)**
4. **Business evaluation, value & recommendations (20%)**
5. **Deployment & production readiness (15%)**
6. **Monitoring & maintenance (15%)**

Are we solving the right problem?

Do we understand the data well enough to solve it?

Does the analytical solution work?

Is it useful, and what should we do based on it?

Can this become a reliable operational solution?

How do we ensure it keeps working over time?

Problem definition & business understanding (15%)	Data understanding & background (15%)	Methods, analysis & technical evaluation (20%)	Business evaluation, value & recommendations (20%)	Deployment & production readiness (15%)	Monitoring & maintenance (15%)
Clearly defines the problem, context and decision to be supported	Provides relevant domain/background information and motivates the selected data	Clearly describes preprocessing, features, methods and analytical workflow	Interprets the technical results in the context of the original business problem	Proposes a realistic plan for moving from the current analysis/prototype to a usable operational solution	Identifies relevant technical, data/model and business KPIs to monitor
Identifies key stakeholders/users and explains why the problem matters	Describes the dataset, data sources, important variables and their relevance to the problem	Selects appropriate methods and explains why they are suitable for the problem and data	Evaluates whether the solution creates meaningful business or operational value	Describes the required workflow, data flows, tools, interfaces, infrastructure and dependencies at an appropriate level	Identifies possible data drift, model degradation, system failures or changing business conditions
Defines measurable success criteria, relevant business KPIs and important constraints	Evaluates data quality, missing values, biases, limitations and important assumptions	Uses suitable baselines, validation strategy and technical evaluation metrics	Relates improvements over baselines/current practice to business consequences, costs, benefits or risks	Explains how the solution should be tested and validated before operational use, including relevant data, functional, integration or end-to-end testing	Defines appropriate alerts/thresholds and explains who should respond when something goes wrong
Clearly scopes what is and is not addressed in the project	Demonstrates meaningful exploratory analysis and understanding of patterns relevant to the task	Presents and compares results clearly; discusses errors, failure cases, robustness and limitations of the analysis	Provides specific, evidence-based recommendations: deploy, pilot, iterate, collect more data, or stop	Addresses reproducibility, versioning, reliability, fallback/rollback, security/privacy, scalability and staged introduction where relevant	Describes maintenance actions such as retraining, recalibration, review, rollback or retirement of the solution

Suggested report structure

- **Problem definition and business understanding** – Describe the context, problem/opportunity, stakeholders, decision to be supported, success criteria and important constraints. *(2–3 pages)*

- **Data understanding and background** – Present relevant domain background and describe the data, including sources, important variables, quality issues, assumptions and limitations. *(3–4 pages)*

- **Methods, analysis and technical evaluation** – Describe data preparation, methods/models, baselines, validation approach and evaluation metrics. Present and compare the main technical results and limitations. *(4–6 pages)*

- **Business evaluation, value and recommendations** – Interpret the results in relation to the original problem. Discuss business/operational value, relevant KPIs, costs, risks and trade-offs, and provide evidence-based recommendations. *(3–5 pages)*

- **Deployment and production readiness** – Describe how the analysis/prototype could become a reliable usable solution. Outline the workflow, required components, testing/validation, and relevant considerations such as reproducibility, versioning, security, scalability or staged deployment. *(3–5 pages)*

- **Monitoring and maintenance** – Describe how the solution should be monitored and maintained. Identify relevant data, solution-performance and business KPIs, possible failures, alerts, responsible stakeholders and appropriate maintenance actions. *(2–4 pages)*

Page ranges are indicative. Focus on clarity, relevance and evidence rather than length.

FAQs- Project (Group)

What is the ideal group size? 5-6 people.

Which tools/technique to use for analysis: You can select any software/method/tool you think is best suited.

Do we need to be present physically for the report submission? No, a PDF report needs to be submitted on Canvas.

When will the groups be assigned?: One week after the deadline of registration.

Grading map

Letter grade	Range
A	100% to 90%
B	< 90% to 80%
C	< 80% to 70%
D	< 70% to 60%
E	< 60% to 50%
F	< 50%

Assignment 1 grade [20%] + Assignment 2 grade [20%] + Group project grade [60%] -> Final grade

Next to come

In the next lecture (16th September), we will start with the core concepts of the lifecycle of a data science project.

In the second part of the lecture, we will look into the Individual assignment 1.

Dataset presentations



Questions & Discussion

Nisha Dalal

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